# INFO-VERSE

## Software Engineering Diagrams

*InfoVerse — A Digital Encyclopedia*

Prepared for Software Engineering documentation using Sparx Enterprise Architect as the modeling reference.

### Project Overview

InfoVerse is a digital encyclopedia website where users can register and log in, search and read articles, save useful content, maintain reading history, and provide feedback. Administrators manage users, articles, categories, and content.

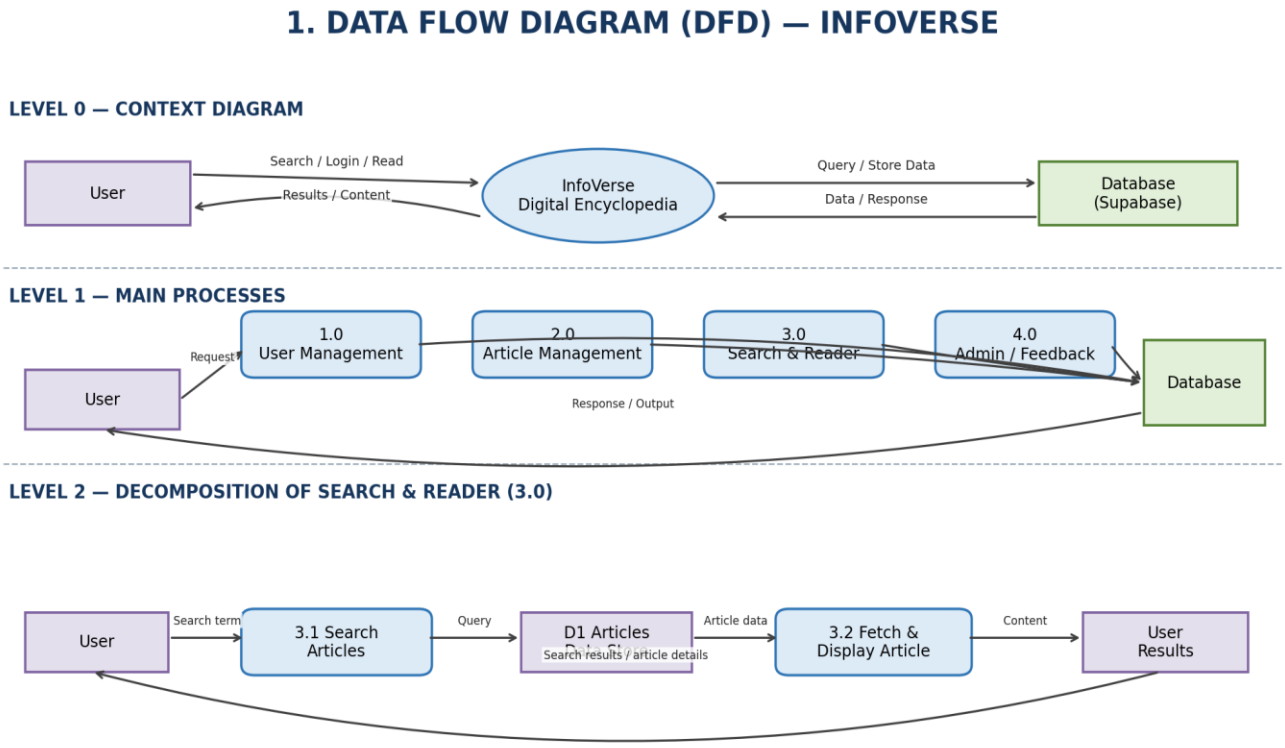
### Diagrams Included

1. Data Flow Diagram (DFD) — Level 0, Level 1, and Level 2
2. UML Use Case Diagram
3. UML Class Diagram
4. UML Sequence Diagram — User Login
5. Entity–Relationship (ER) Diagram

**Note:** These diagrams are supplied as images in this Word document. To submit a native Sparx model, recreate the shapes and connectors in Enterprise Architect and save the project as an .qea file.

## 1. Data Flow Diagram (DFD)

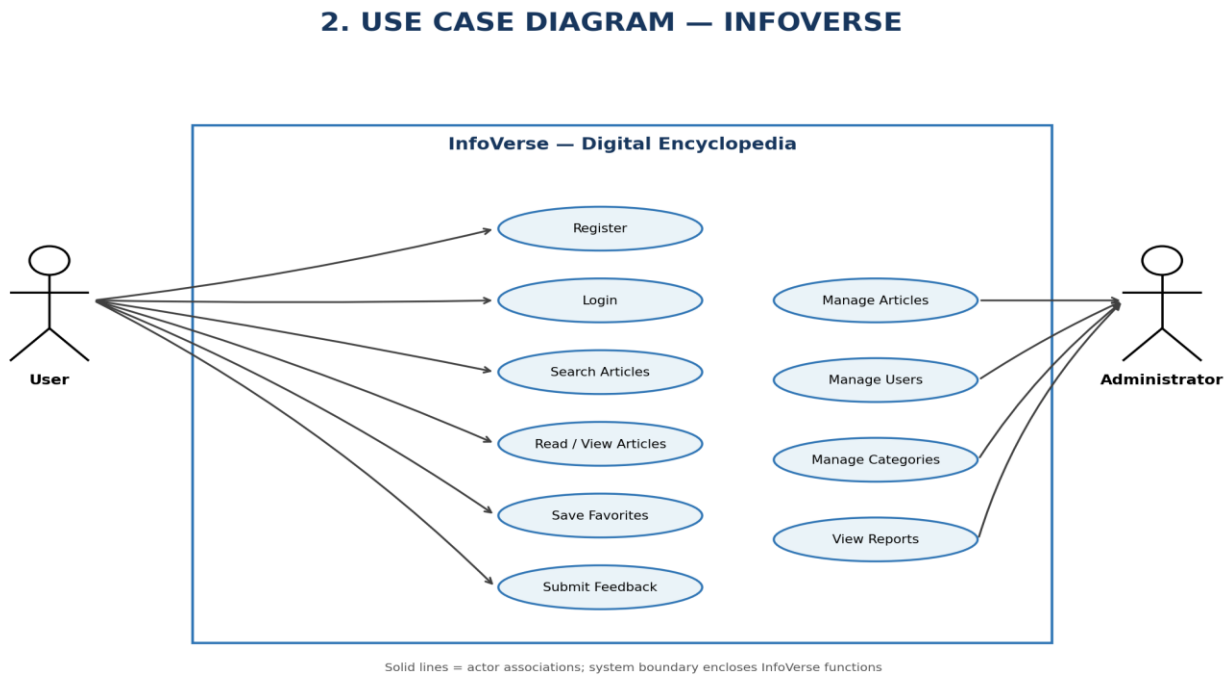
Shows the context of InfoVerse, its major processes, and a decomposed search/article-reading process. The DFD includes three levels in one figure.




DFD notation note: external entities are rectangles, processes are rounded rectangles/circles, data stores are labeled storage shapes, and arrows are labeled with data flows.

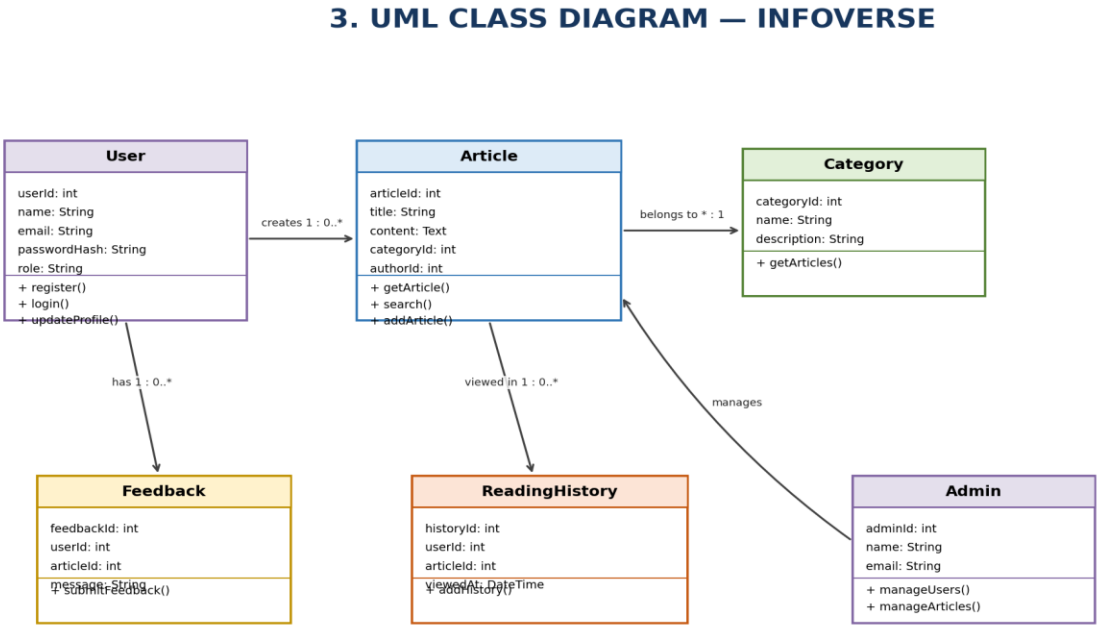
## 2. UML Use Case Diagram

Shows the main services available to users and the content-management tasks available to the administrator.



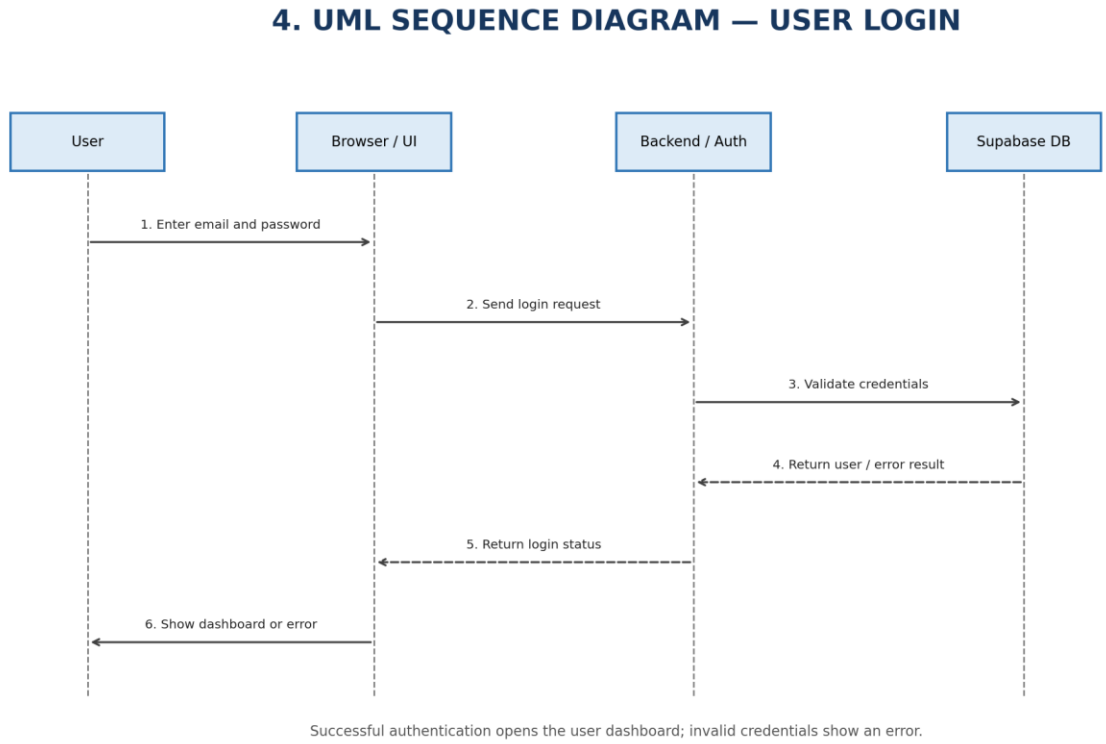

## 3. UML Class Diagram

Shows the main object classes, example attributes and operations, and relationships between users, articles, categories, feedback, reading history, and administrators.



## 4. UML Sequence Diagram — User Login

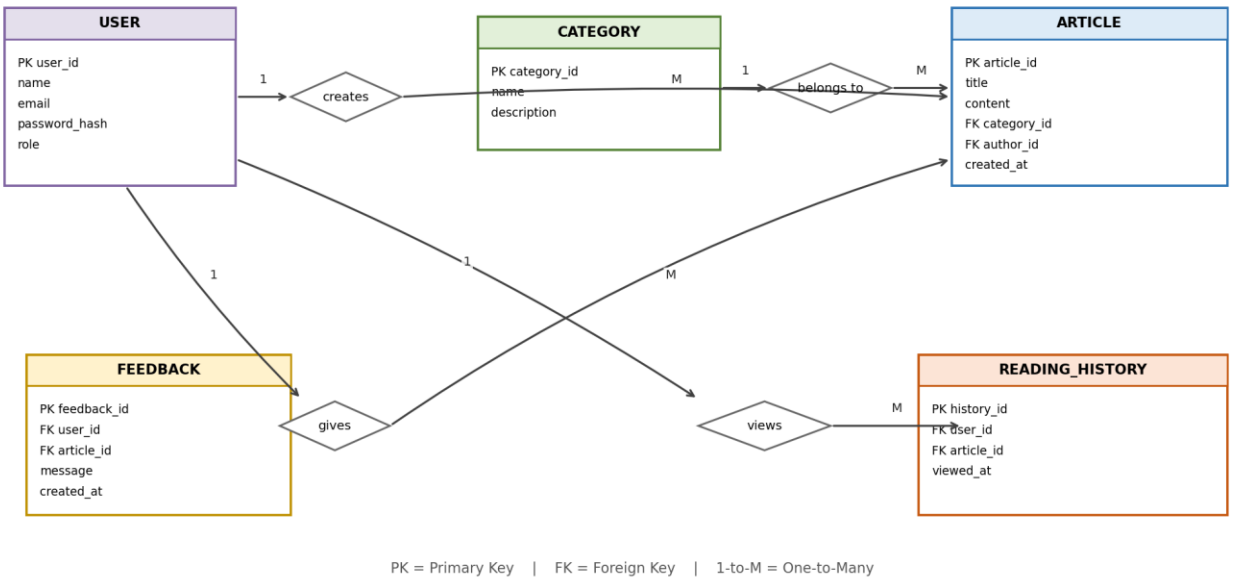
Shows the chronological interaction between the user, browser interface, authentication backend, and Supabase database during login.




## 5. Entity–Relationship (ER) Diagram

Shows the main data entities and example primary-key/foreign-key relationships for the InfoVerse database.

## 5. ENTITY-RELATIONSHIP (ER) DIAGRAM — INFOVERSE




PK = Primary Key | FK = Foreign Key | 1-to-M = One-to-Many

## Creating These Diagrams in Sparx Enterprise Architect

1. Open Enterprise Architect and open or create your InfoVerse project.
2. Create a package named “InfoVerse Diagrams” in the Browser panel.
3. Right-click the package and choose Add Diagram. Create one diagram for each of the five types.
4. For the Use Case, Class, and Sequence diagrams, select the matching UML diagram type and use the Toolbox elements.
5. For the DFD, use a suitable Data Flow/Flowchart notation available in your installed version. If DFD is not listed, use a suitable diagram and clearly label the DFD levels.
6. For the ER diagram, use the database/ERD notation available in your Sparx version; define primary keys, foreign keys, and relationship cardinalities.
7. Add labeled connectors, keep the layout readable, and export each diagram from Enterprise Architect as an image or PDF when required.

## Modeling Assumptions

The diagrams are conceptual documentation based on the InfoVerse features described for this project. Verify that the actual implementation and Supabase schema use the same entities, fields, and authentication flow before final submission.